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Investor Presentation | Japan

Japan Summer School: Semiconductor Production Equipment

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SEMICONDUCTOR PRODUCTION EQUIPMENT
Japan
Industry View **Attractive**

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Japan Summer School 2026



Morgan Stanley

July 21, 2026

INVESTOR PRESENTATION

Japan Summer School

Semiconductor Production Equipment

In this presentation, we discuss industry basics as well as related companies.

MORGAN STANLEY RESEARCH
Japan

MORGAN STANLEY MUFG Securities., Ltd.

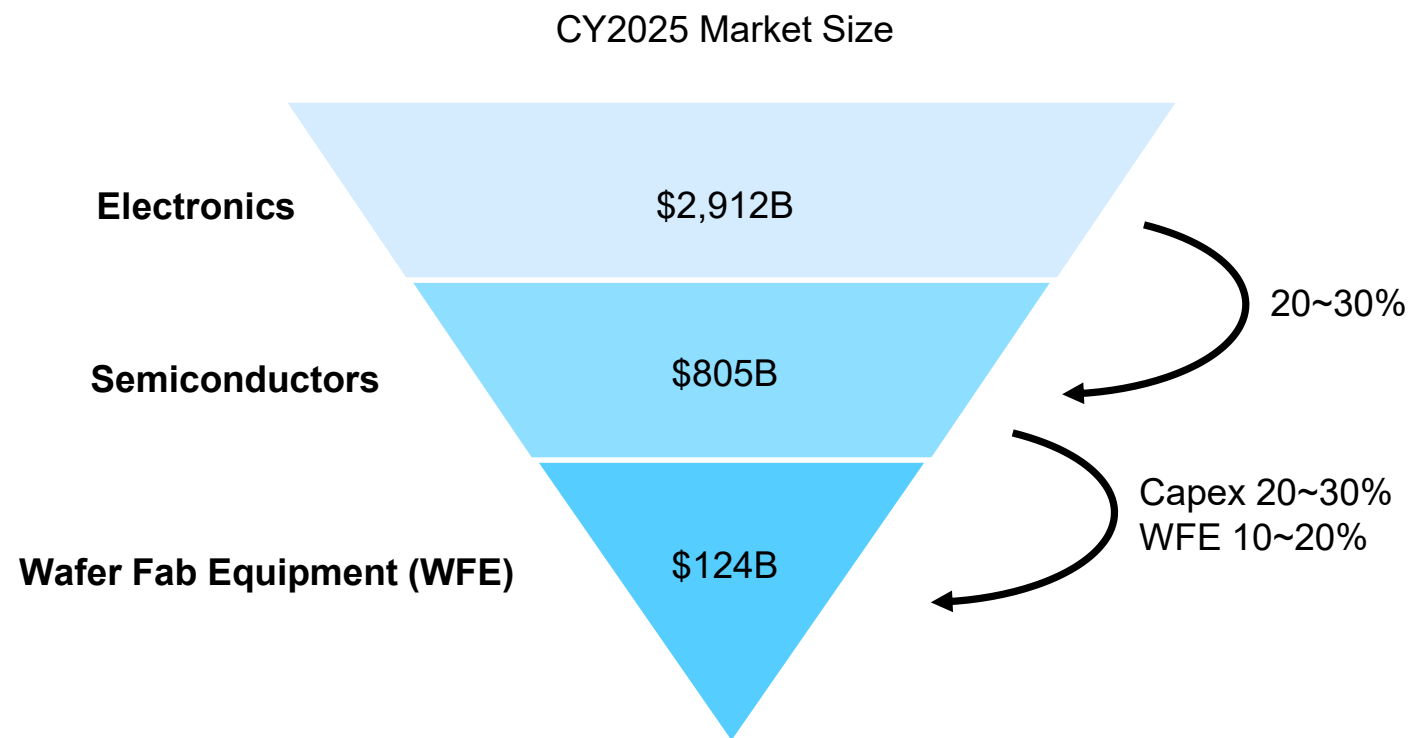
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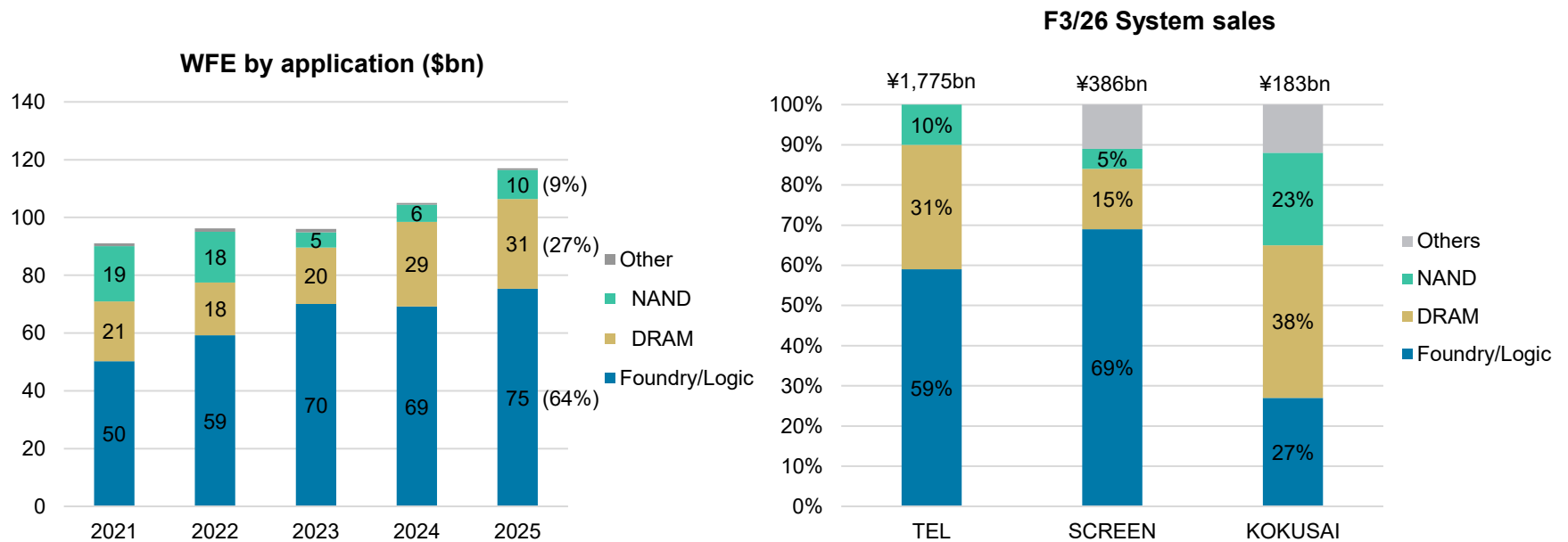
Semiconductor Equipment Market Size



Source: Gartner, Tokyo Electron, Morgan Stanley Research

WFE Market by Application

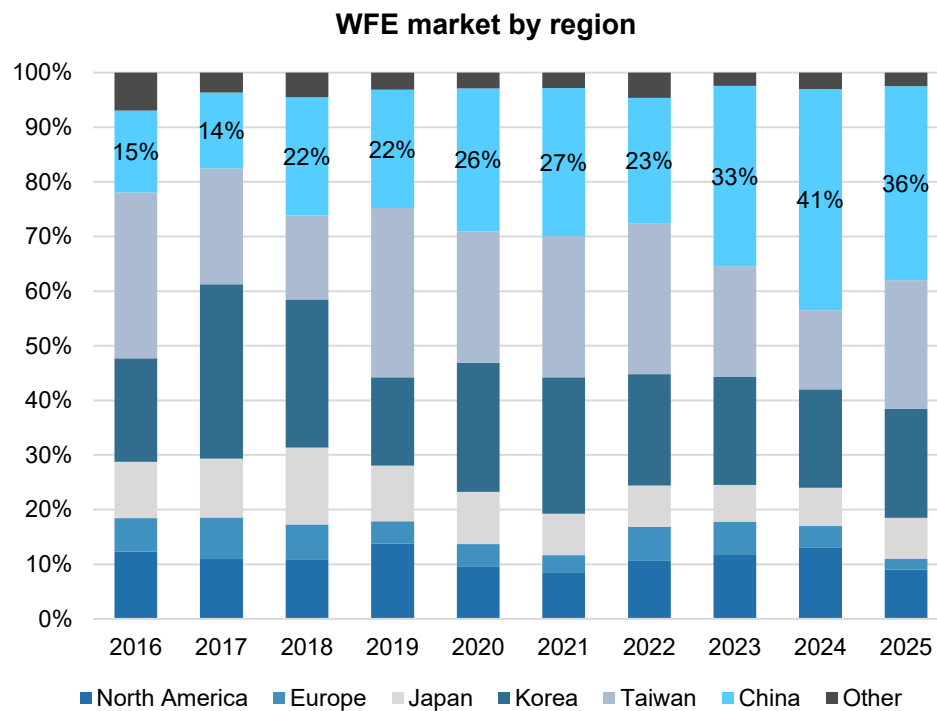
By application, Logic/Foundry is the largest segment; within Logic/Foundry, TSMC accounts for roughly 30–50%
Among Japanese equipment makers, SCREEN is typically viewed as a Logic/Foundry play, TEL as a DRAM-related play, and KOKUSAI as a NAND-related play



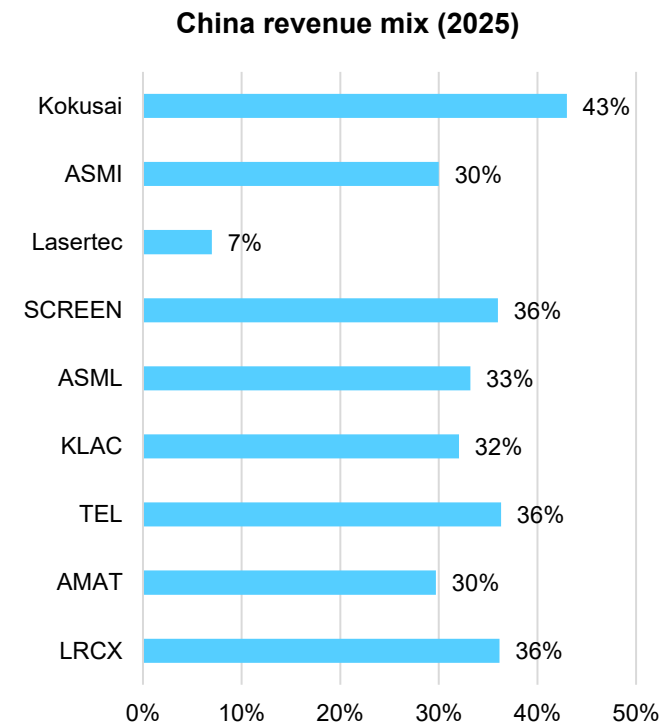
Source: Morgan Stanley Research

WFE Market by Region

China's market has expanded rapidly over the past decade; major front-end equipment makers' China exposure is broadly in line with the overall WFE market



Source: Morgan Stanley Research



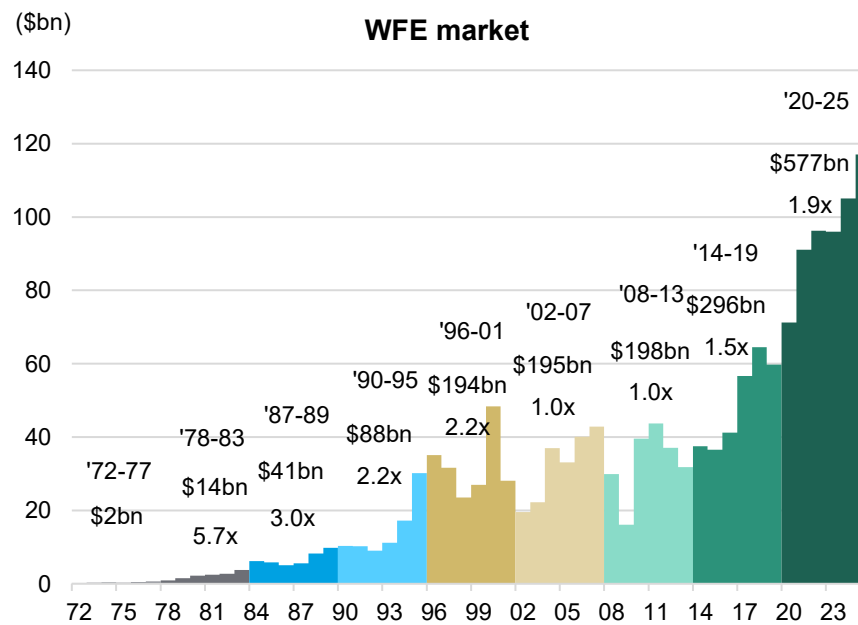
Source: Company data, Morgan Stanley Research

The Silicon Cycle

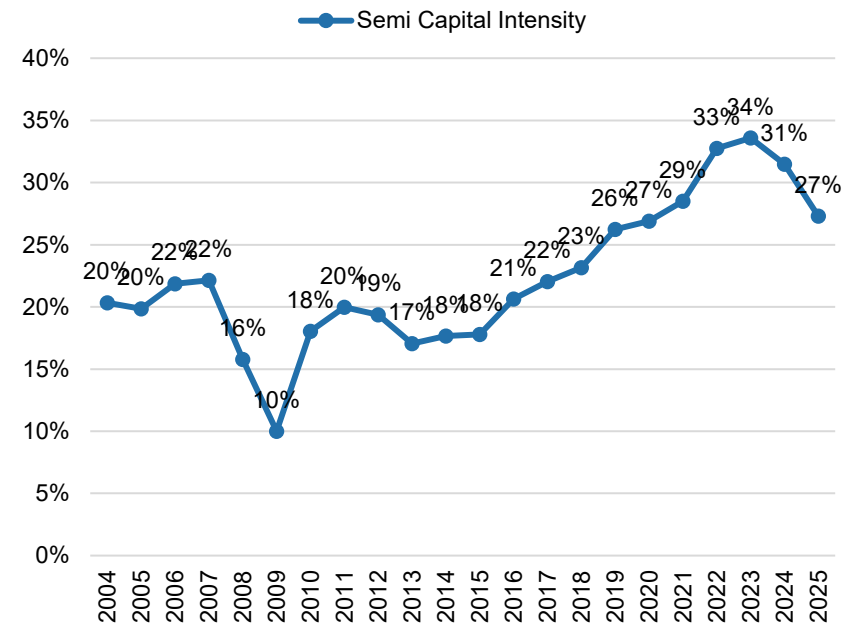
The equipment market stagnated after the dot-com bubble, but returned to growth following the late-2010s expansion in data-center investment.

Excluding the post-2000 stagnation, the market has continued to roughly double every six years on a cumulative basis.

Capital intensity (semiconductor capex / sales) has risen from around 20% historically to roughly 30% today.

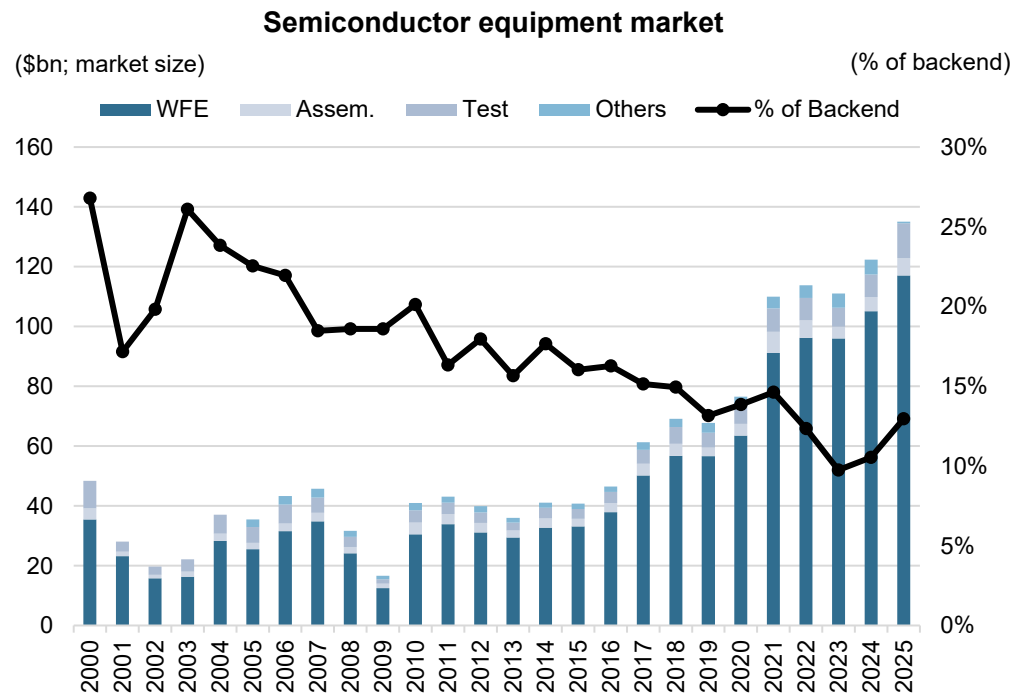


Source: Gartner, Morgan Stanley Research

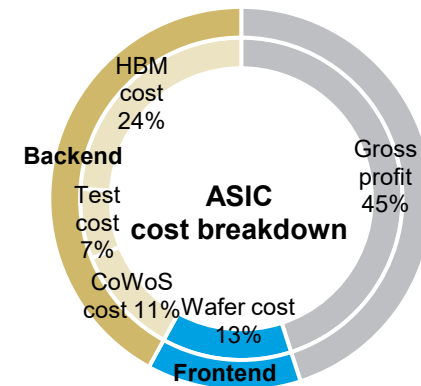
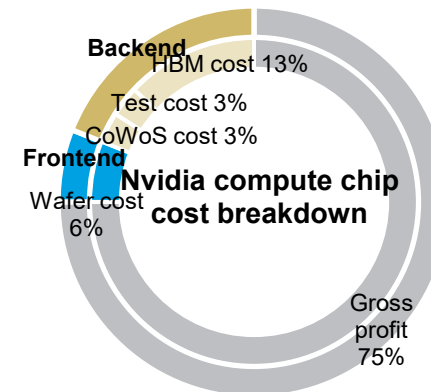


Semiconductor Equipment Market: Front-End vs. Back-End

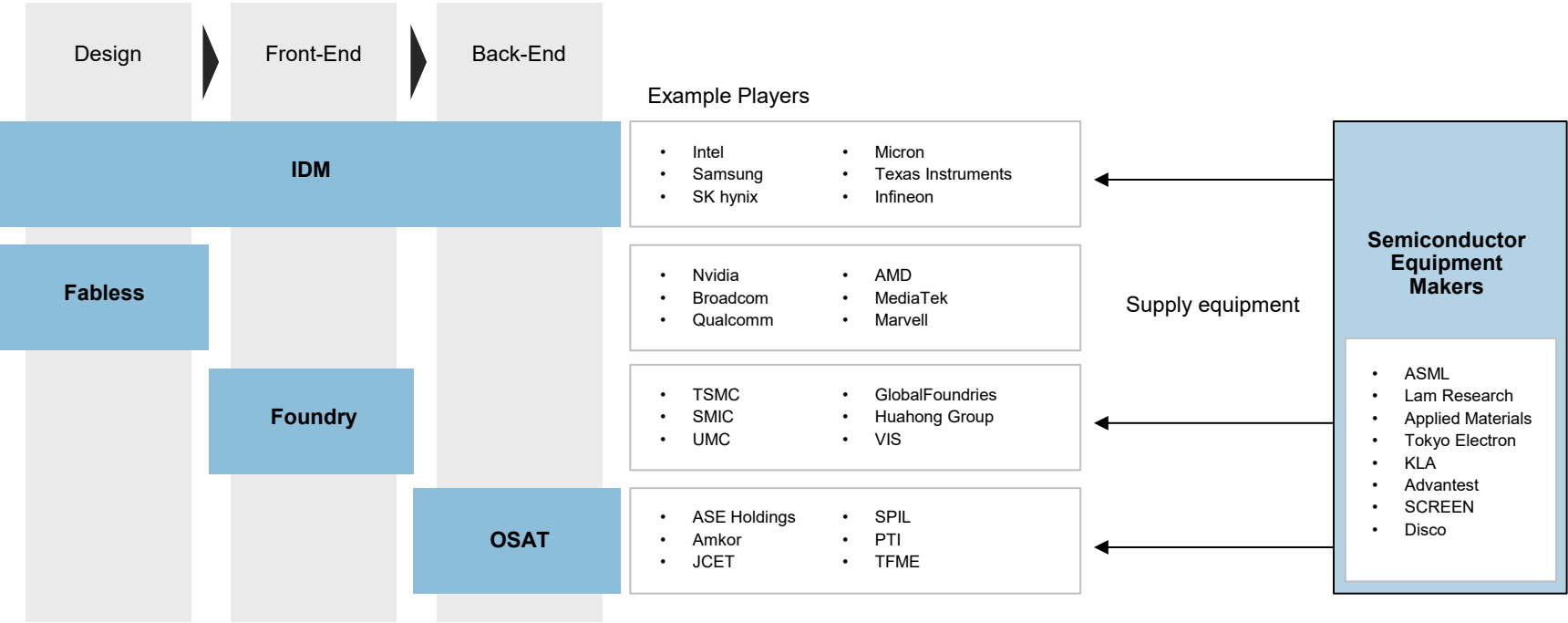
The equipment market splits roughly 9:1 between front-end and back-end. As front-end tools' value-add and unit prices have risen, the back-end share trended down for over 20 years, but has recently rebounded amid the surge in the AI semiconductor market.



Source: SEMI, Morgan Stanley Research

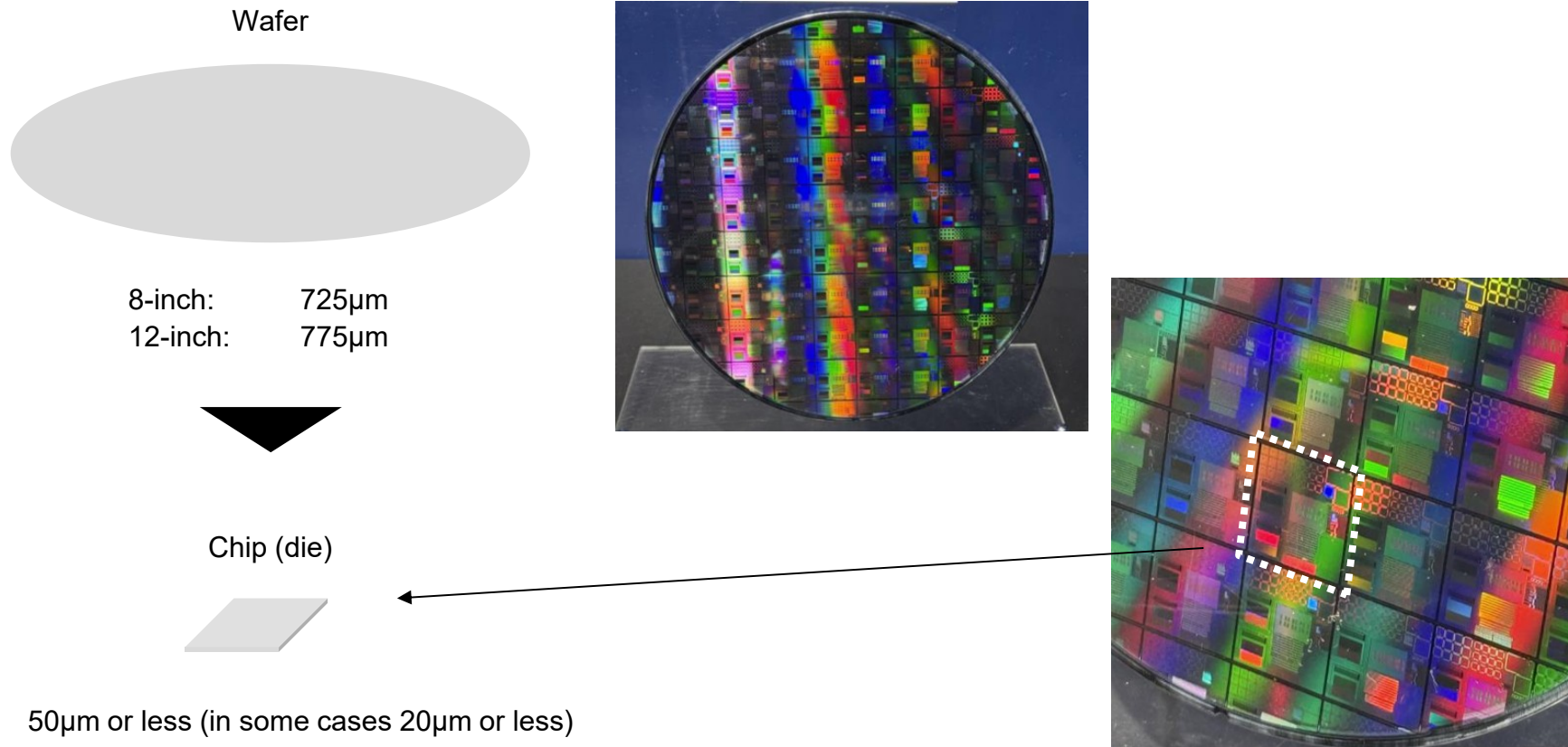


Horizontal Specialization in the Semiconductor Industry



Source: Morgan Stanley Research

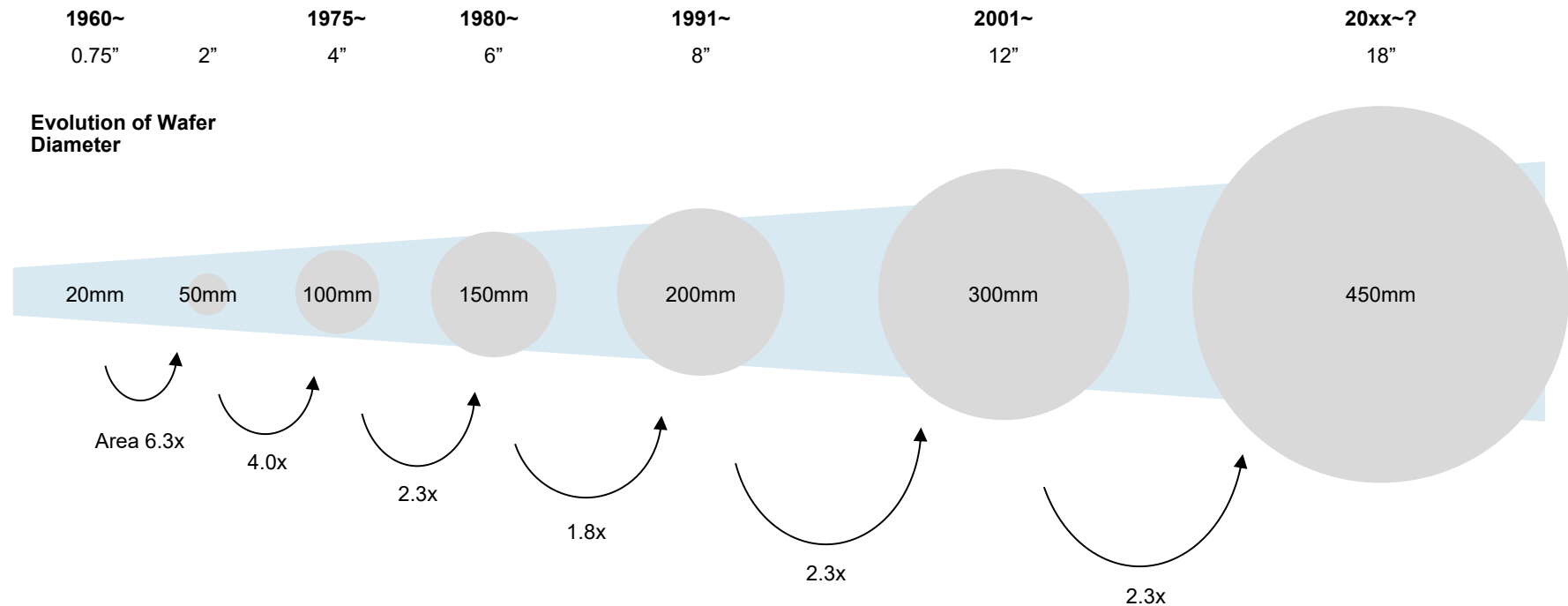
Silicon Wafer



Source: Morgan Stanley Research

History of Wafer Diameter Growth

A larger wafer area yields more chips per wafer, lowering the manufacturing cost per chip

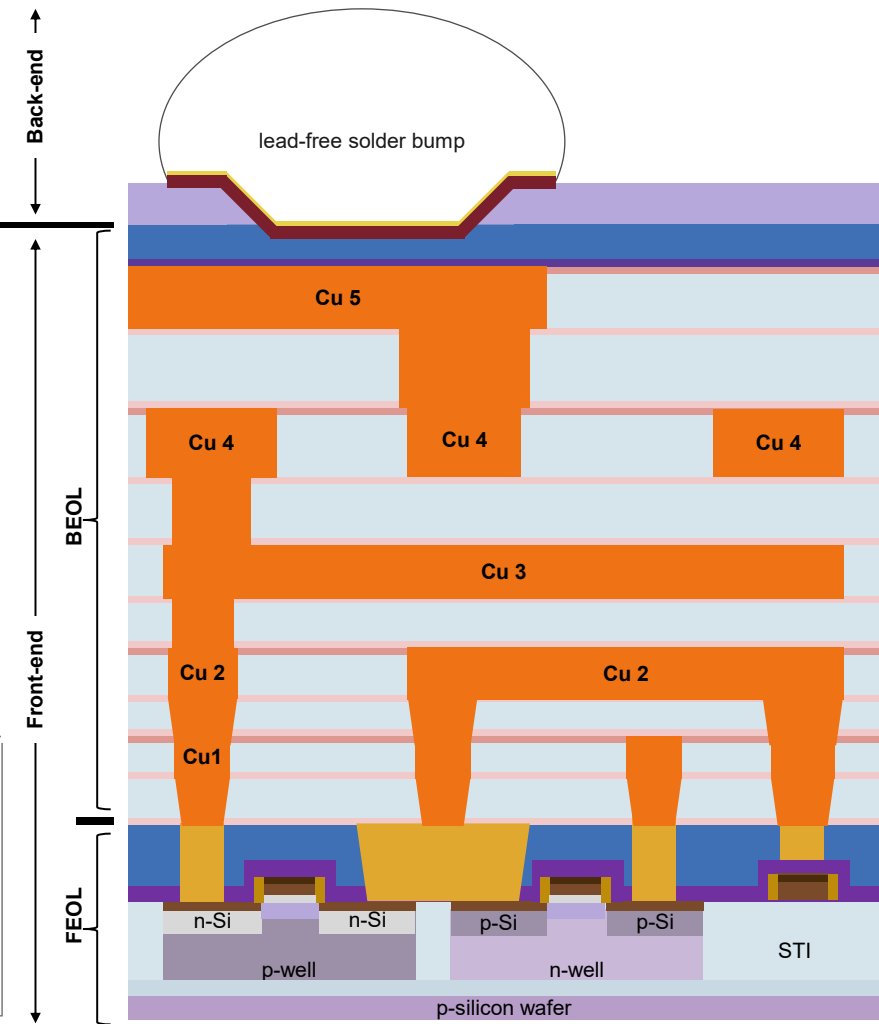
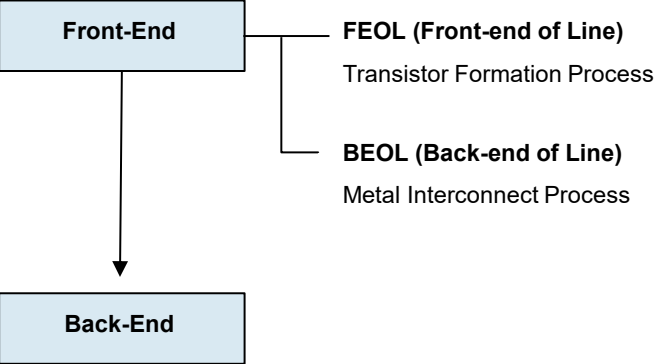


Diameter (mm)	20	50	100	150	200	300	450
Area (mm ²) = ((1/2) x Diameter) ² x π	314	1,963	7,854	17,671	31,416	70,686	159,043

Source: Tokyo Electron, Morgan Stanley Research

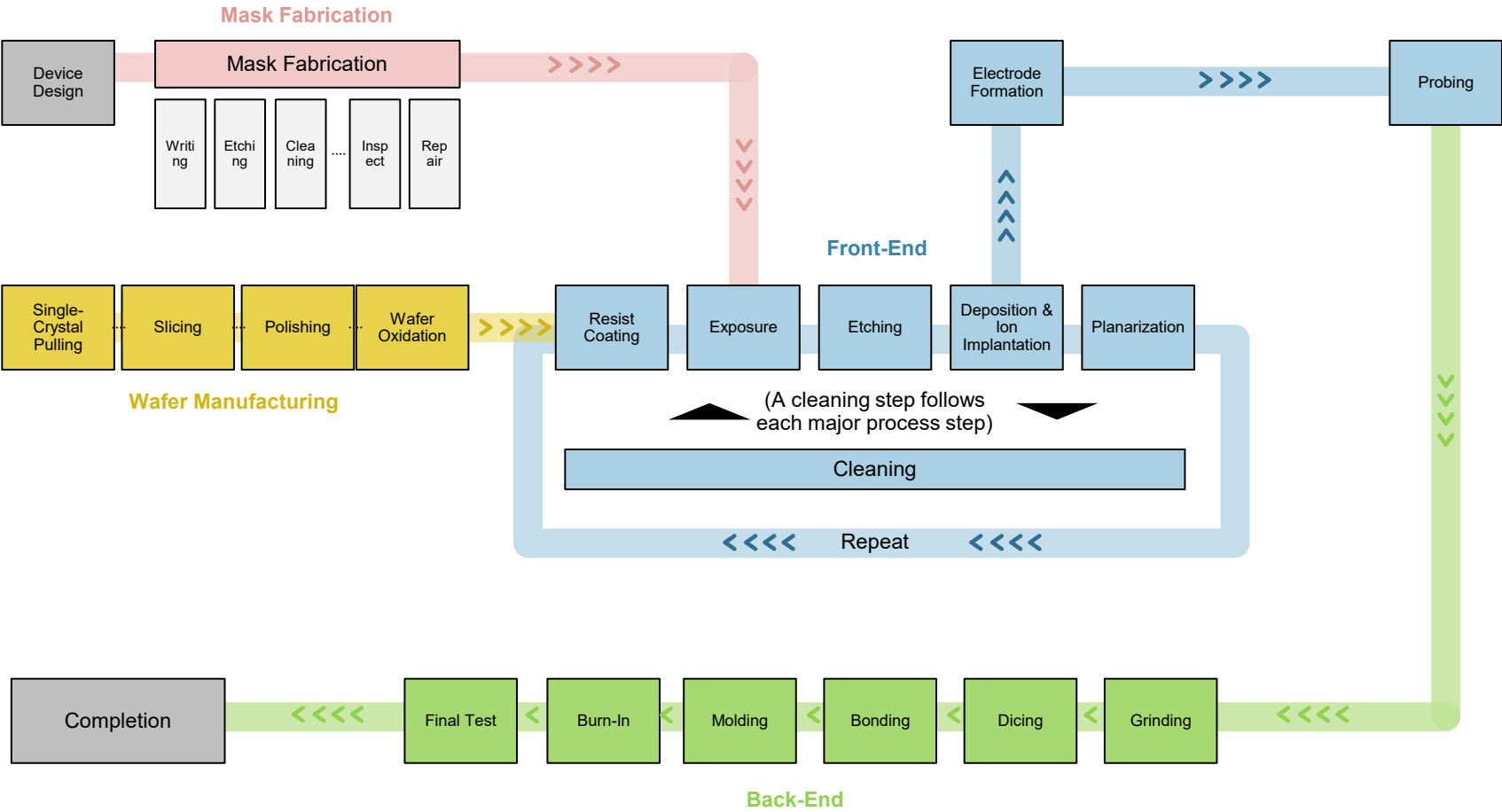
Semiconductor Manufacturing Process Overview

Semiconductor Manufacturing Process



Source: Morgan Stanley Research

Semiconductor Manufacturing Flow



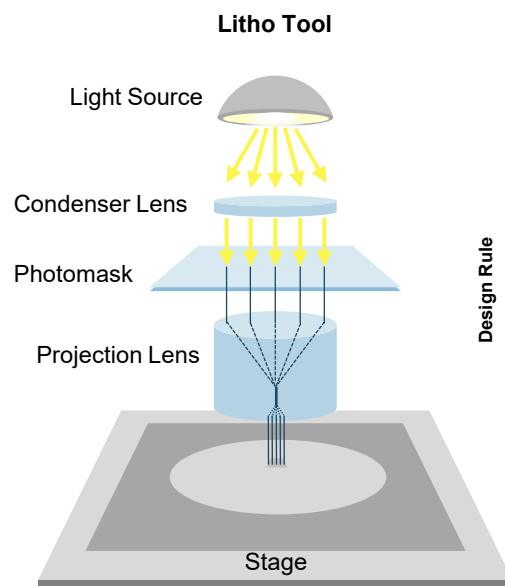
Source: SEAJ, Morgan Stanley Research

Lithography Process

Light is projected through a patterned photomask to transfer the mask pattern onto the photoresist coated on the wafer surface

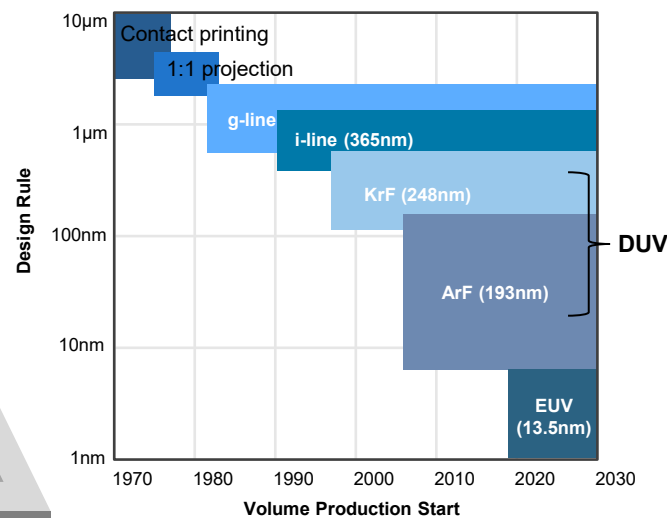
Key equipment and major vendors

- Litho tools — ASML, Nikon, Canon
- Coater/developers — Tokyo Electron

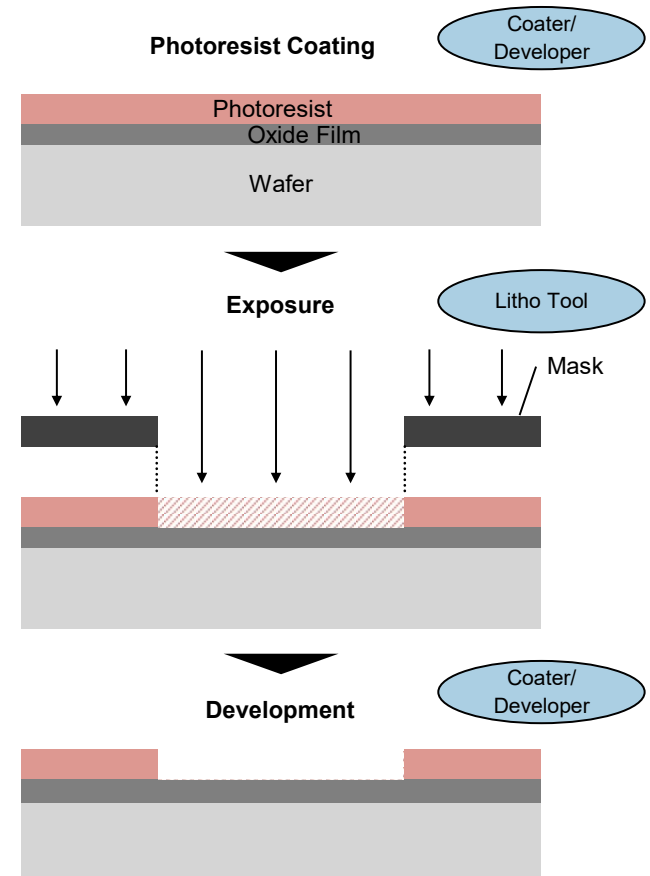


Source: Morgan Stanley Research

History of shorter-wavelength exposure light sources as feature sizes shrink

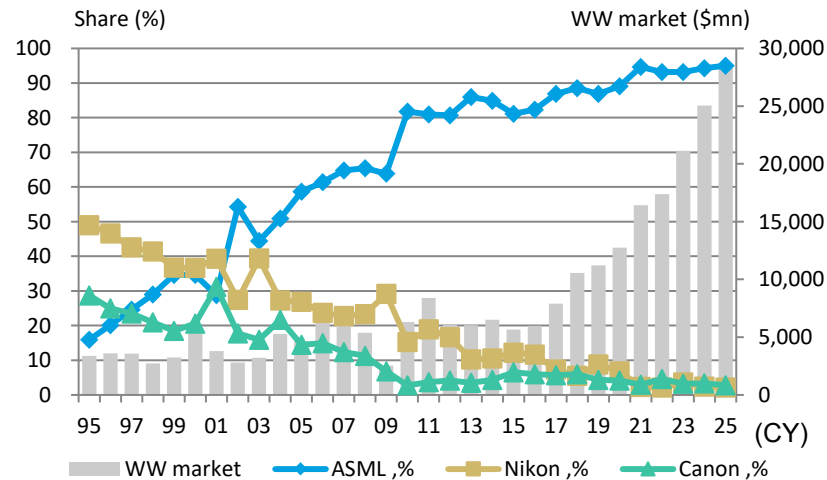
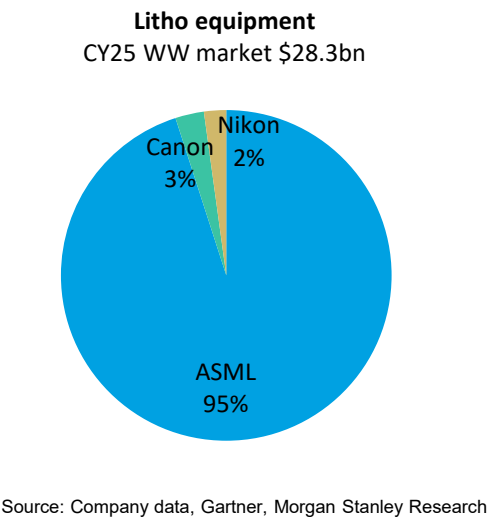
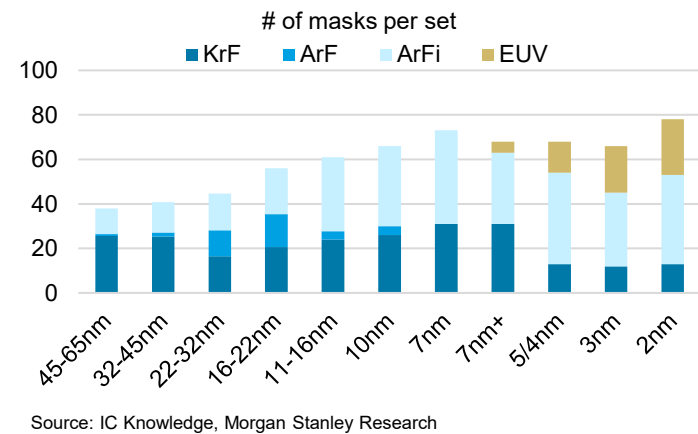
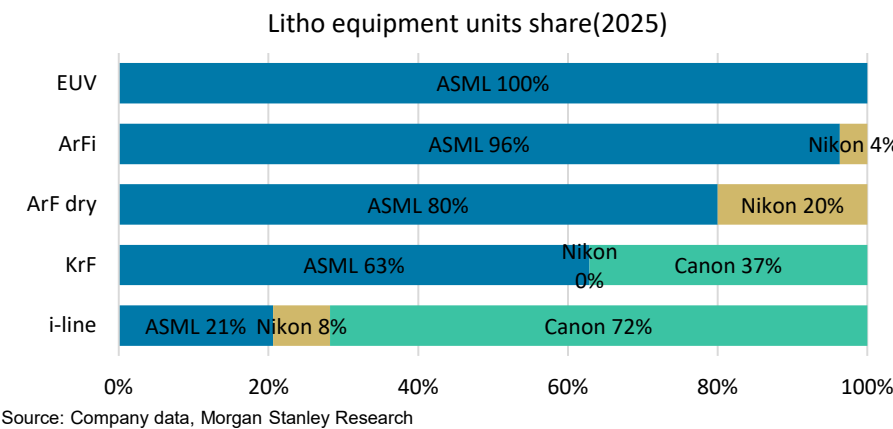


Source: Tokyo Electron, Morgan Stanley Research

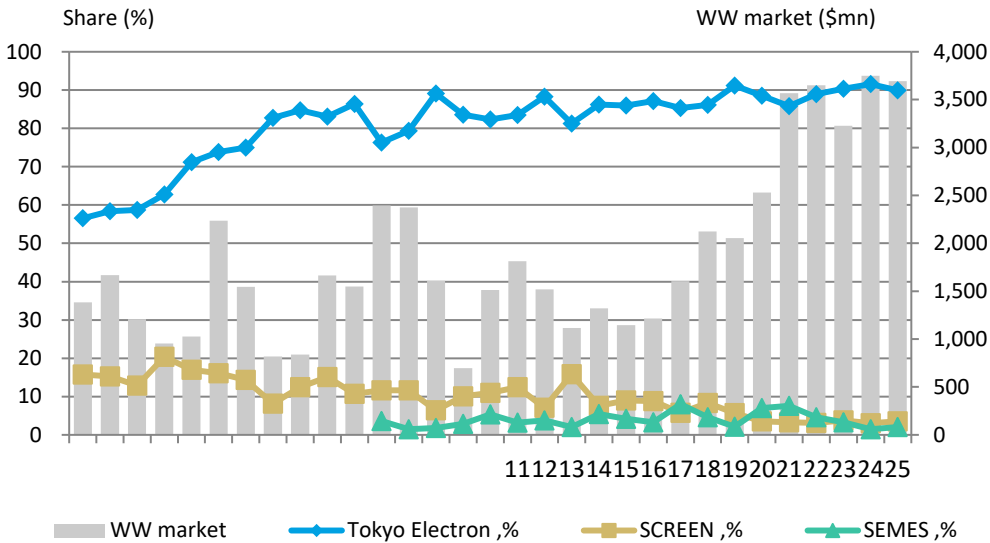
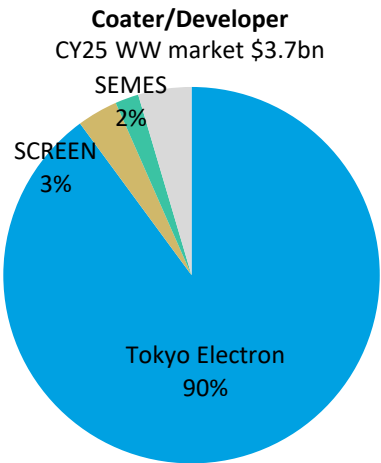


Source: Morgan Stanley Research

Lithography Equipment Market Share (2025)



Coater/Developer Market Share (2025)



Source: Company data, Gartner, Morgan Stanley Research

Etching Process

Uses mainly plasma to remove material from the substrate or thin film through physical and chemical action, forming fine relief patterns on the substrate

Key equipment/materials and major vendors

- Etchers — Lam Research, Tokyo Electron, Applied Materials
 - RF power supplies — Advanced Energy, MKS Instruments, Daihen
 - Electrostatic chucks — Shinko Electric, NGK, TOTO, Niterra, Kyocera

RIE (Reactive Ion Etching) equipment

Gas inlet

- Etching gas
- Inert gas (noble gases, etc.)

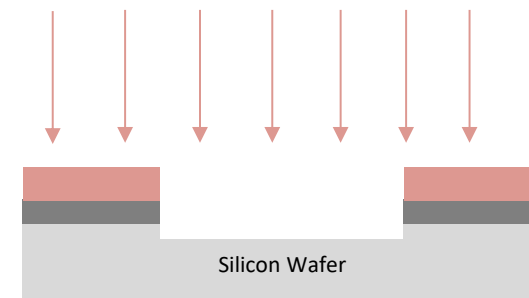
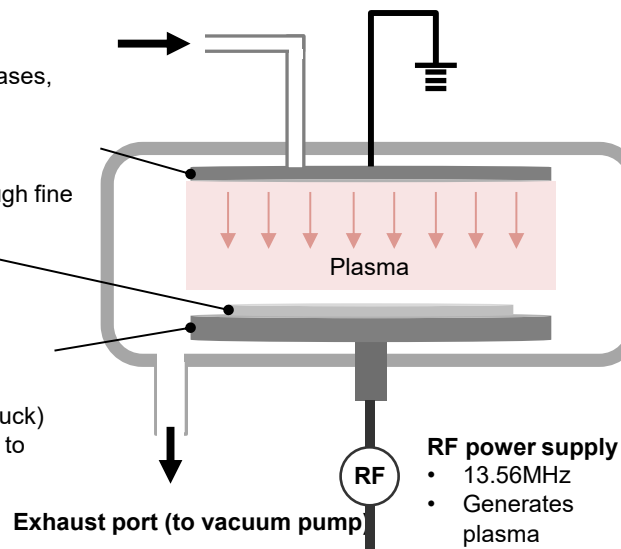
Upper electrode / showerhead

- Ejects gas through fine holes

Wafer

Lower electrode (chuck)

- Holds the wafer (electrostatic chuck)
- Applies RF bias to accelerate ions



Source: Morgan Stanley Research

Dry Etch Equipment Market Share (2025)

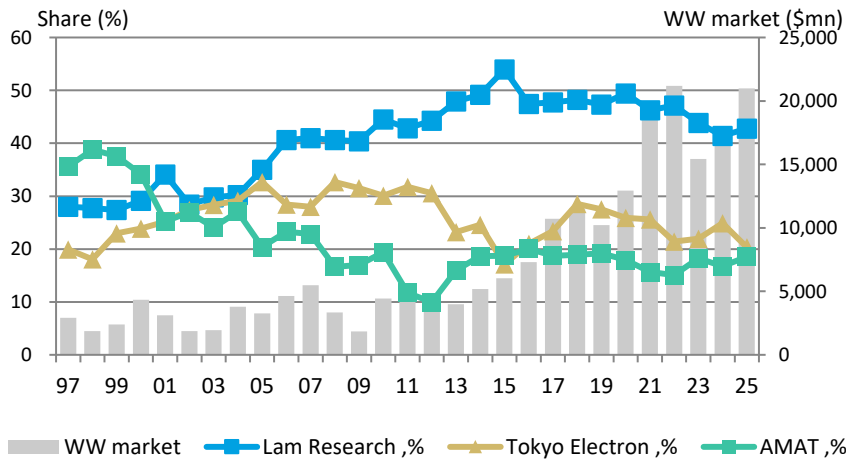
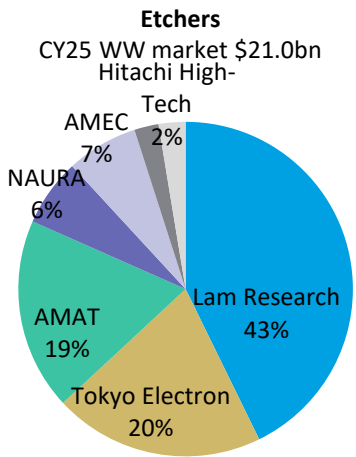
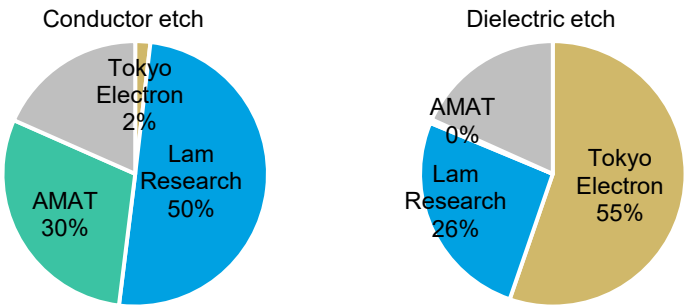
- Wet Etching

Dry Etching

Uses chemical solutions; the equipment is similar to cleaning tools
The current mainstream; over 90% of etching is dry, using mainly plasma
- Conductor Etch

Dielectric Etch
- Mainly etching of metals and silicon
Used to form interconnect patterns, gate electrodes, etc.
Etching of silicon oxide, silicon nitride, low-k materials, etc.
Used to form contact holes and the trenches into which Cu interconnects are embedded

Lam/AMAT are strong in conductor etch, while TEL is strong in dielectric etch
Lam dominates dielectric HARC etch for memory holes



Source: Company data, Gartner, Morgan Stanley Research

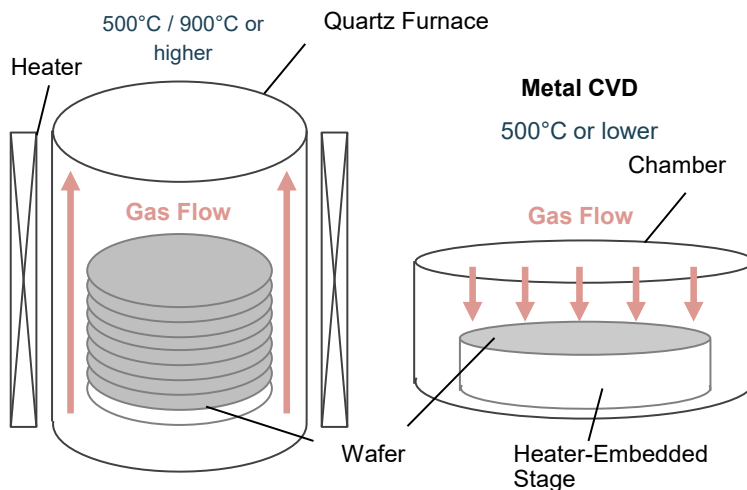
Deposition Process

Forms interconnects and insulating films on the silicon substrate; a variety of deposition tools exist depending on the source gas and deposition temperature

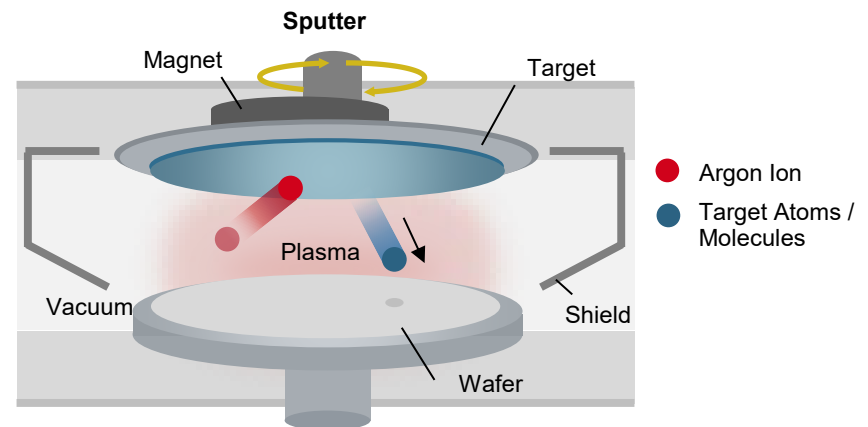
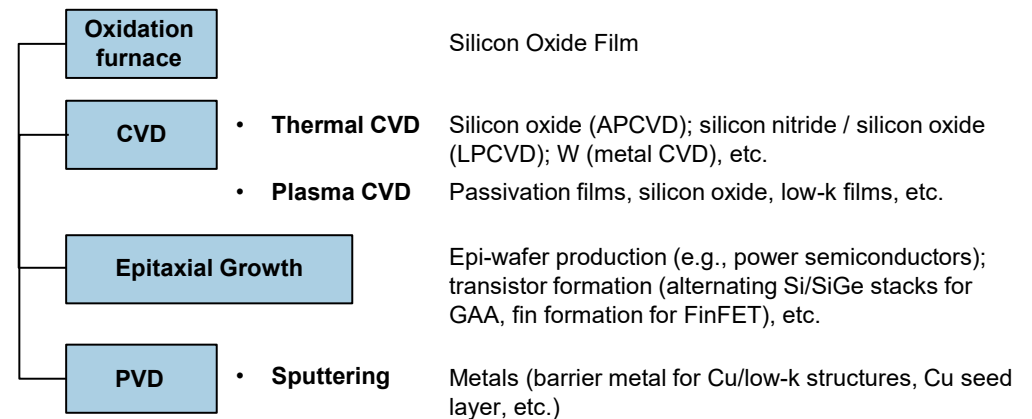
Key equipment and major vendors

- Oxidation furnaces — Tokyo Electron, Kokusai Electric
- Thermal CVD — Kokusai Electric, Tokyo Electron
- Plasma CVD — Applied Materials, Lam Research
- Epitaxy systems— Applied Materials
- Sputters — Applied Materials

Thermal CVD / oxidation furnace (oxidation furnace: no source gas)

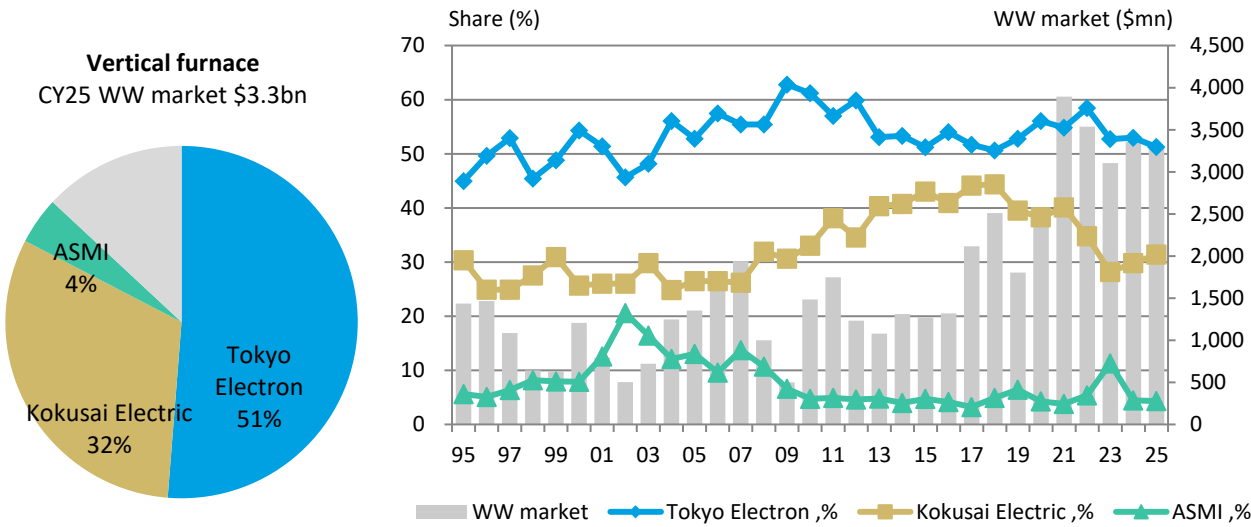


Source: Morgan Stanley Research



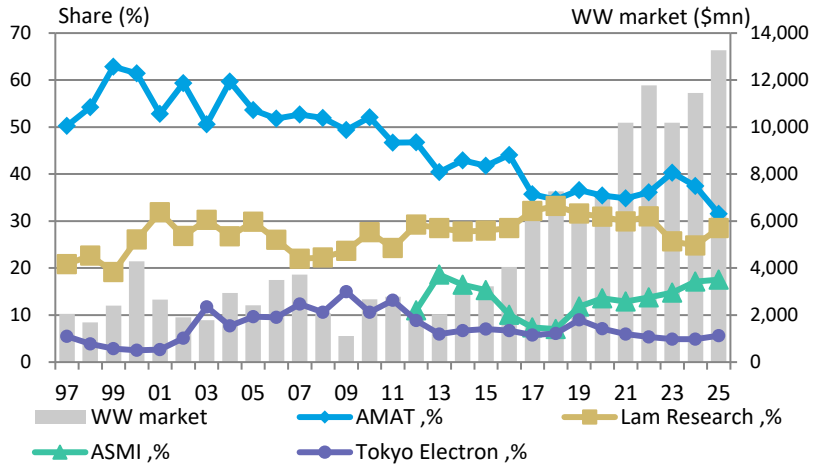
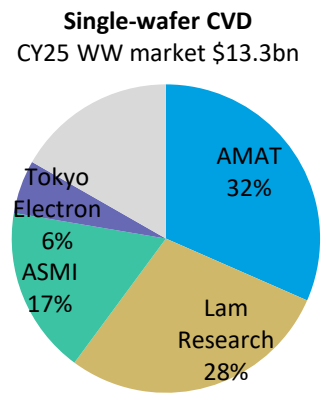
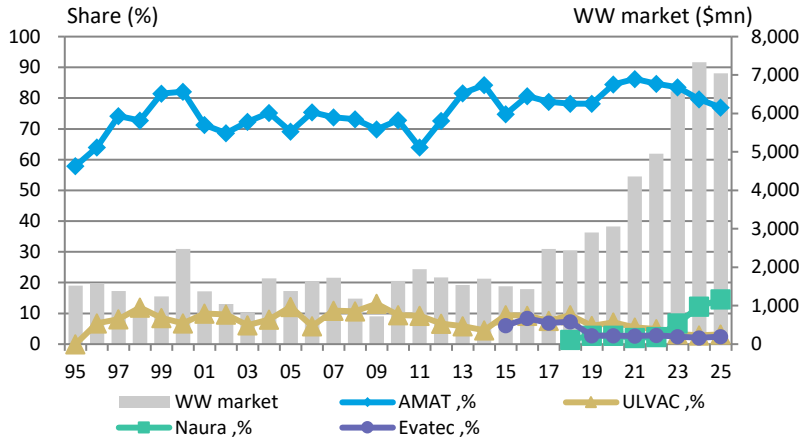
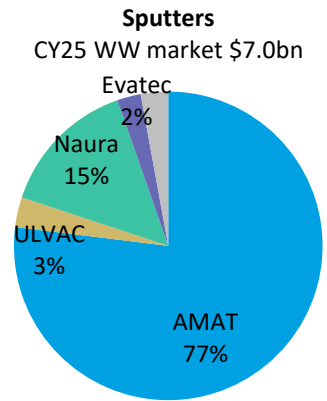
Vertical Thermal-Processing Equipment Market Share (2025)

Oxidation/diffusion furnaces and batch CVD; the equipment structures are largely the same



Source: Company data, Gartner, Morgan Stanley Research

PVD & CVD Equipment Market Share (2025)



Source: Company data, Gartner, Morgan Stanley Research

CMP Process

Combines the chemical reaction of slurry chemicals with physical polishing by the pad to planarize wafer-surface topography to nanometer precision

Key equipment and major vendors

- CMP tools — Applied Materials, Ebara

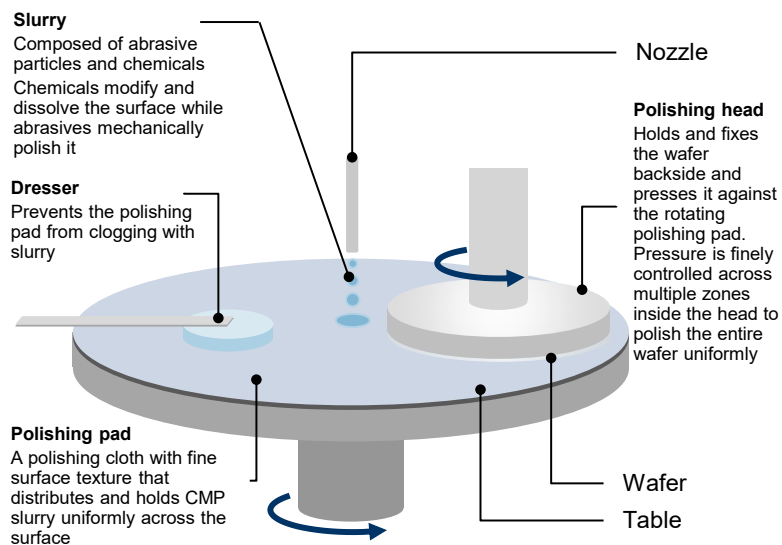
Example CMP applications

Dielectric CMP

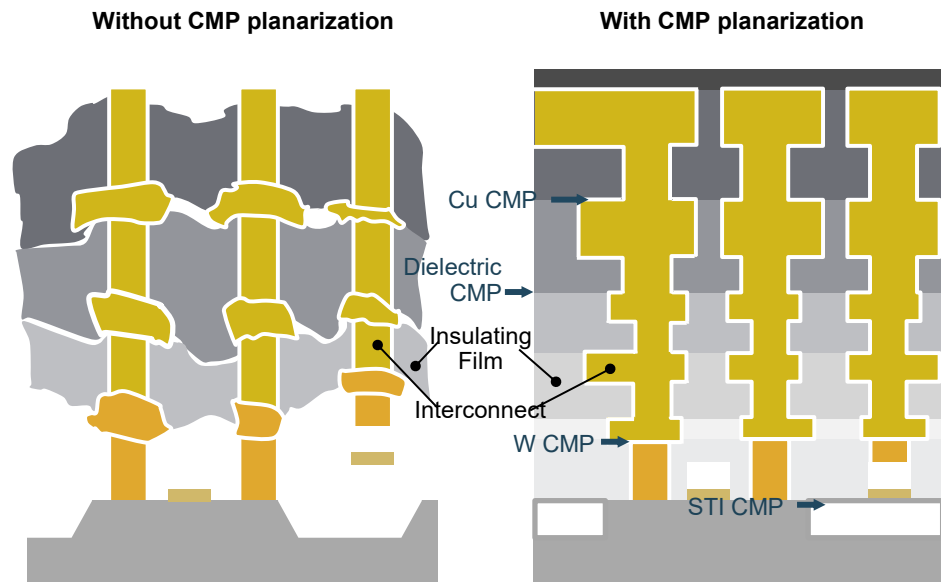
- Non-SiO₂ — low-k
- SiO₂ — interlayer dielectric (ILD), STI

Conductive-film CMP

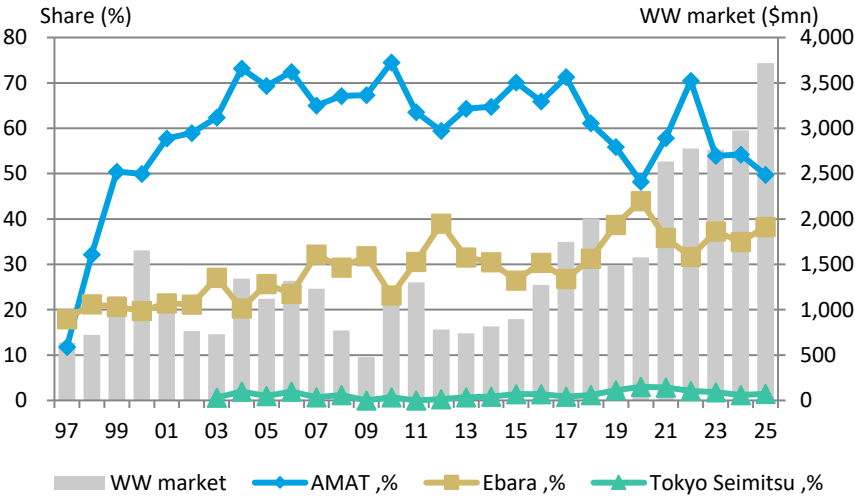
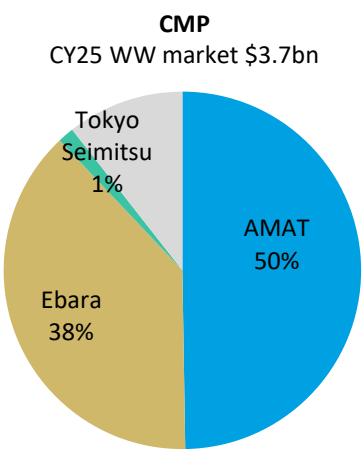
- Conductive-film CMP — tungsten (W) plugs, Cu (damascene process)



Source: Morgan Stanley Research



CMP Equipment Market Share (2025)



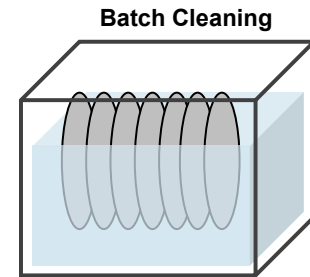
Source: Company data, Gartner, Morgan Stanley Research

Cleaning Process

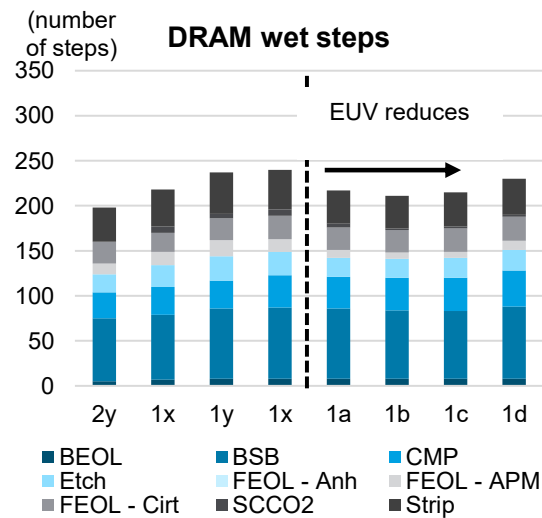
Removes fine particles, metallic contamination, organics, and native oxide from the wafer surface. Performed before and after every other process step, it is said to account for around 30% of the entire semiconductor manufacturing process.

Key equipment and major vendors

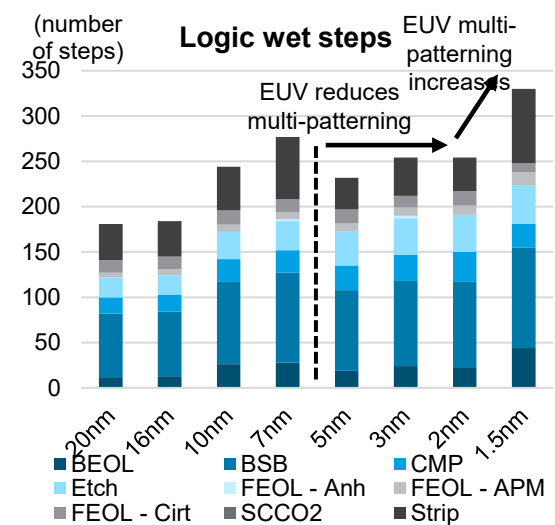
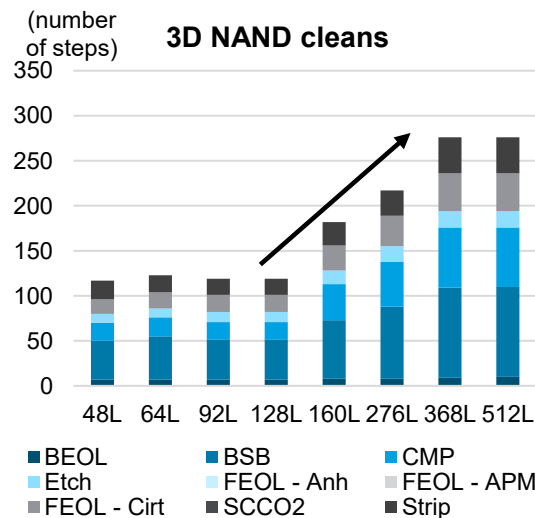
- Single-wafer cleaning tools — SCREEN, Lam Research, Tokyo Electron, SEMES
- Batch cleaning tools — SCREEN, Tokyo Electron



Source: Morgan Stanley Research

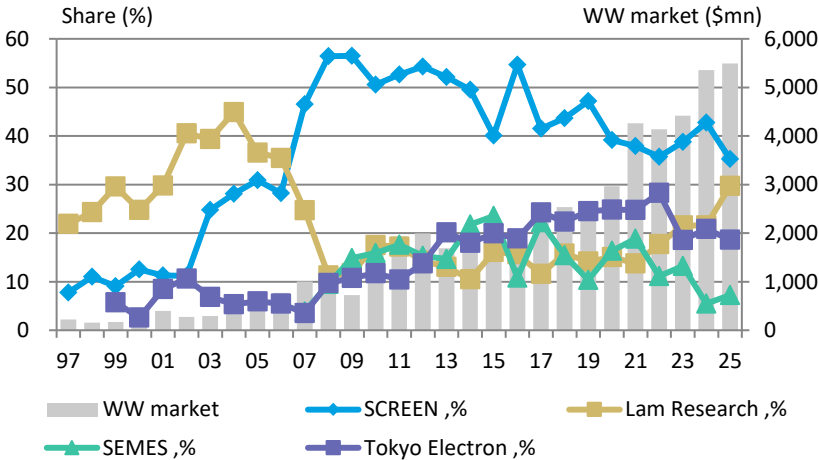
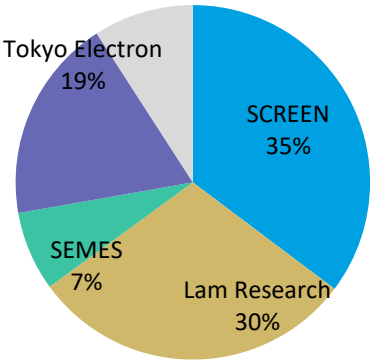


Source: IC Knowledge (SPCC 2021), Morgan Stanley Research

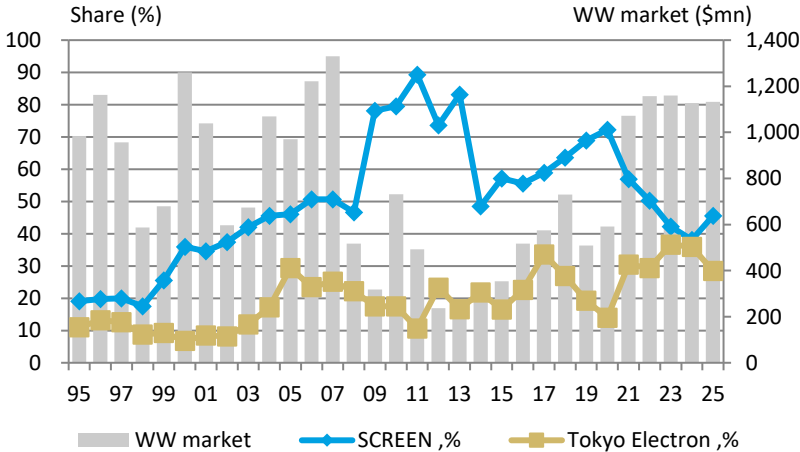
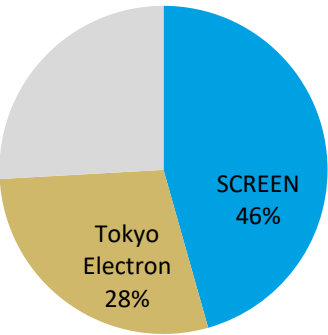


Cleaning Equipment Market Share (2025)

Single-wafer cleaning equipment
CY25 WW market \$5.5bn



Batch cleaning equipment
CY25 WW market \$1.1bn



Source: Company data, Gartner, Morgan Stanley Research

Grinding Process

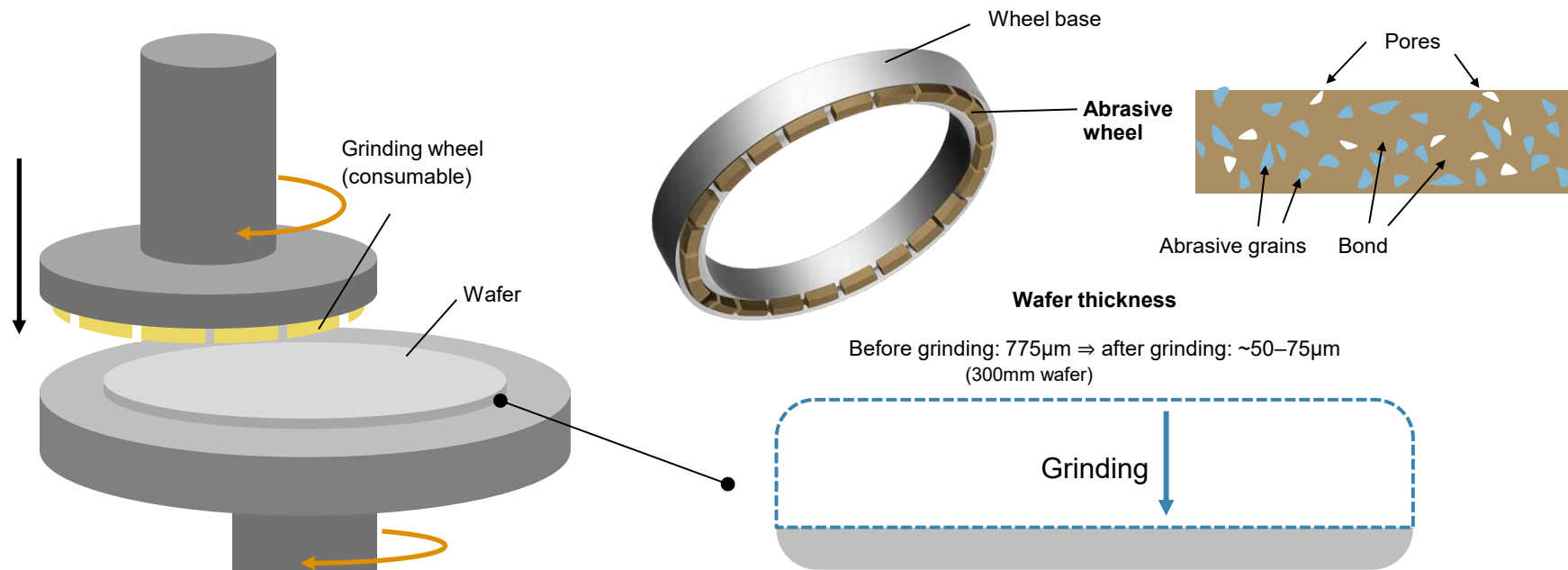
Back-grinding: after circuits are formed on the wafer during the front-end, the circuit-free backside is ground away to thin the wafer

Key equipment and major vendors

- Grinders — Disco, Tokyo Seimitsu (Accretech)

Applications

- Surface grinding in wafer making
- Thinning of device wafers (back-grinding)
- Package grinding



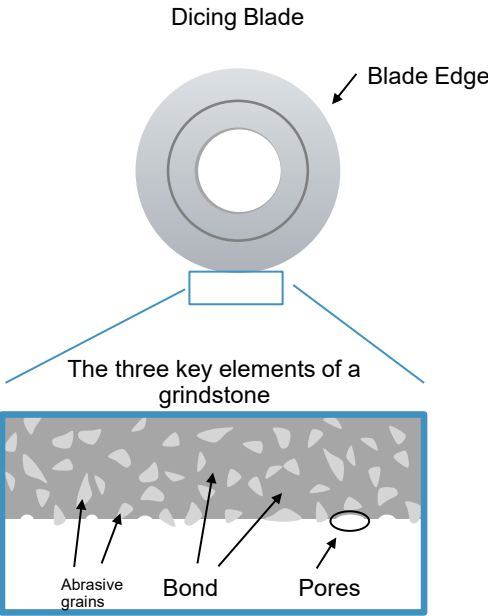
Source: Morgan Stanley Research

Dicing Process

Cuts individual chips (dies) from the wafer — singulation

Key equipment and major vendors

- Dicers — Disco, Tokyo Seimitsu (Accretech)



Source: Morgan Stanley Research

Types of dicing

Blade Dicing		Physically cuts using a dicing blade	Simple, with few process steps
Laser Dicing	Ablation	Irradiates with a laser to instantaneously decompose/sublimate atoms and molecules at the material surface	More process steps & consumables
	Stealth Dicing	Focuses the laser inside the workpiece to form a modified layer, then separates the chips via tape expansion, etc.	Limited range of processable materials
Plasma Dicing		Singulates chips via plasma-based dry etching	More process steps and consumables; requires gas-handling equipment; limited range of compatible materials

Blade dicing

Protective film coating

Laser grooving

Protective film removal

Blade dicing

Stealth Dicing

Tape Remount

Tape expanding

Resist coating

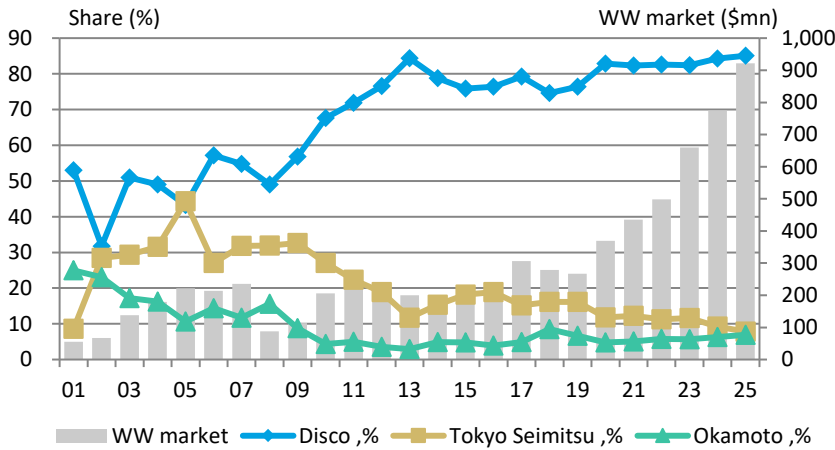
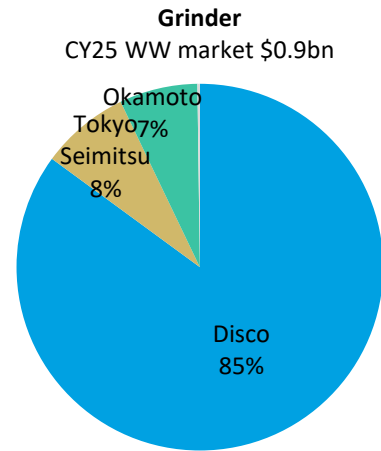
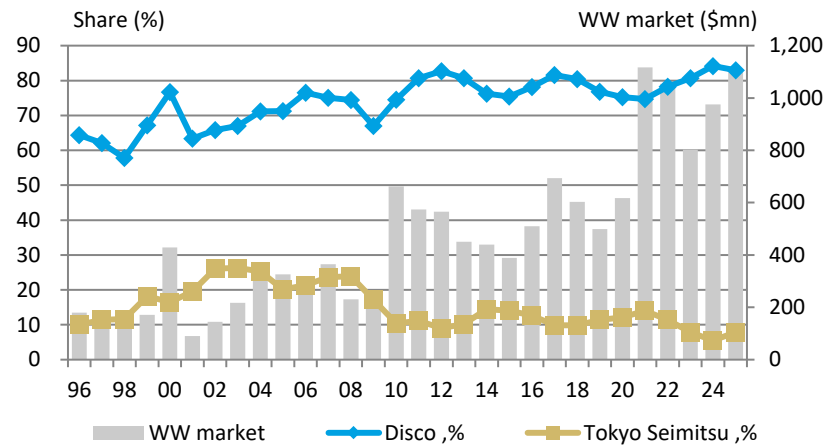
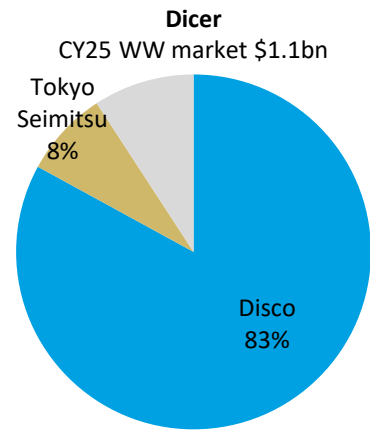
Laser grooving

Remote plasma

Resist removal

Source: Disco, Morgan Stanley Research

Dicer & Grinder Market Share (2025)



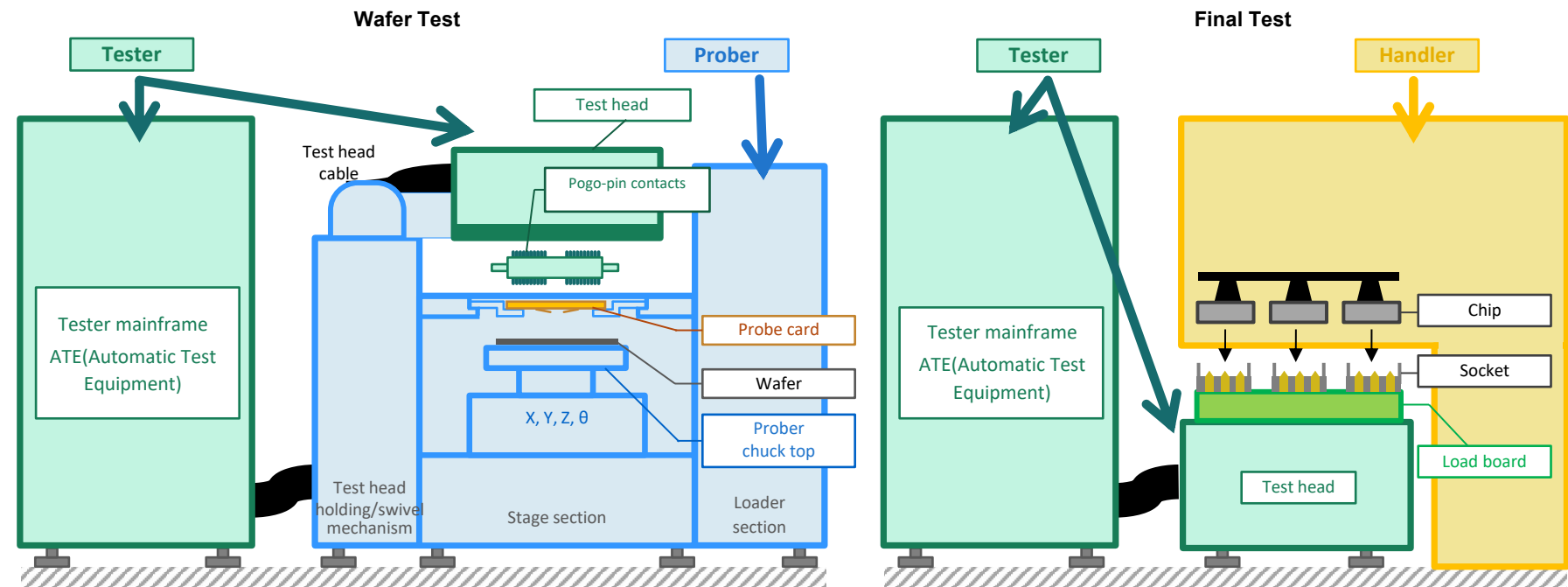
Source: Company data, Gartner, Morgan Stanley Research

Test Process

Verifies that manufactured chips operate as designed and meet high quality and reliability standards. It splits into 'wafer test,' performed at the wafer stage, and 'final test,' performed after packaging

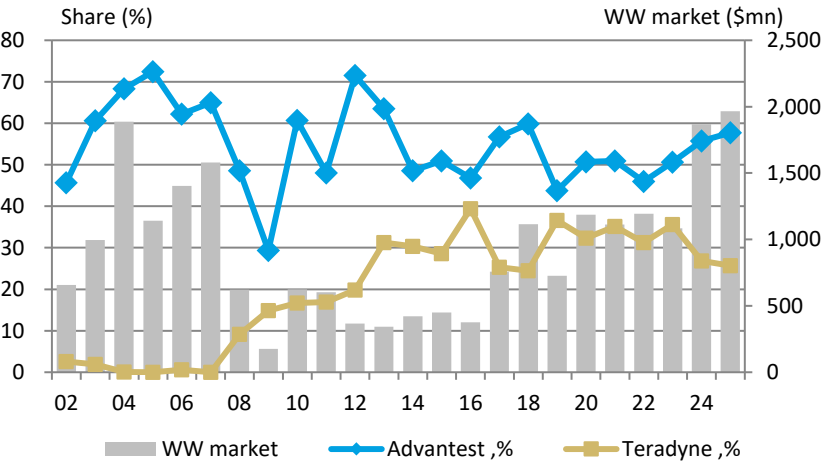
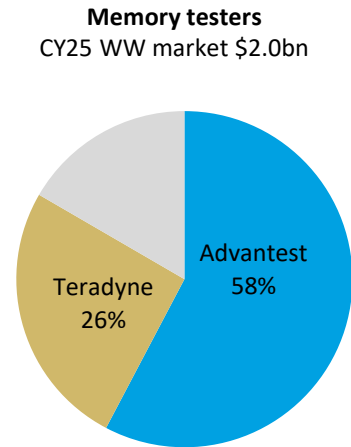
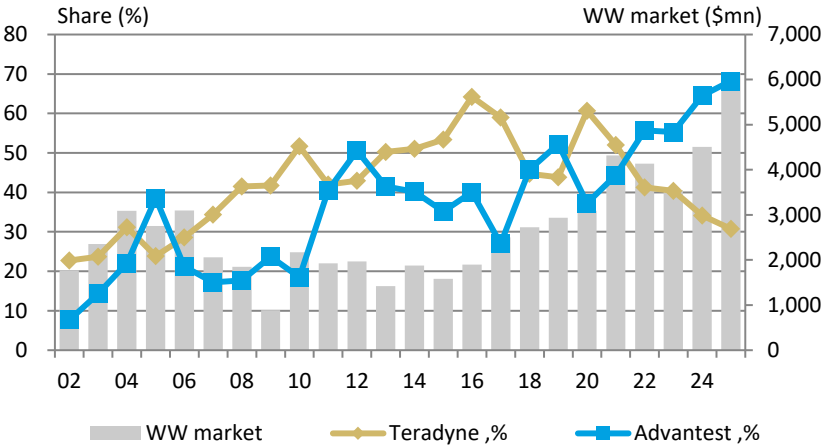
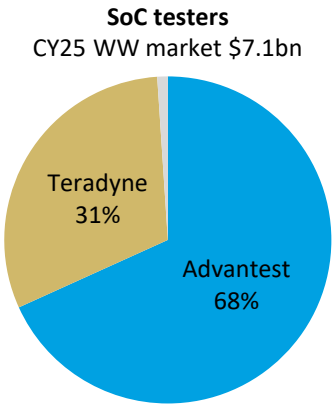
Key equipment/materials and major vendors

- ATE — Advantest, Teradyne
- Probers — Tokyo Electron, Tokyo Seimitsu (Accretech)
- Probe cards — FormFactor, Micronics Japan, Japan Electronic Materials
- Handlers — Hon Precision, Cohu



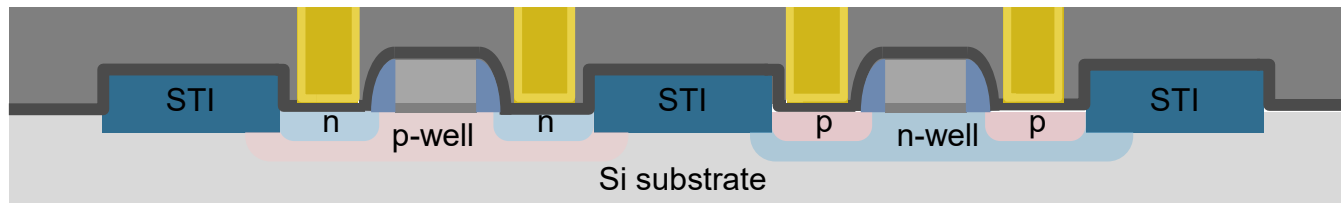
Source: Morgan Stanley Research

Tester Market Share (2025)



Source: Company data, Gartner, Morgan Stanley Research

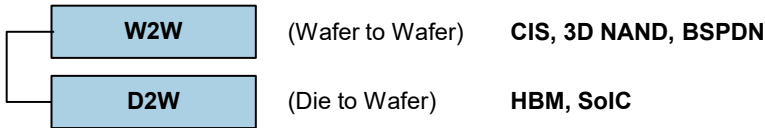
FEOL — CMOS



Source: Morgan Stanley Research

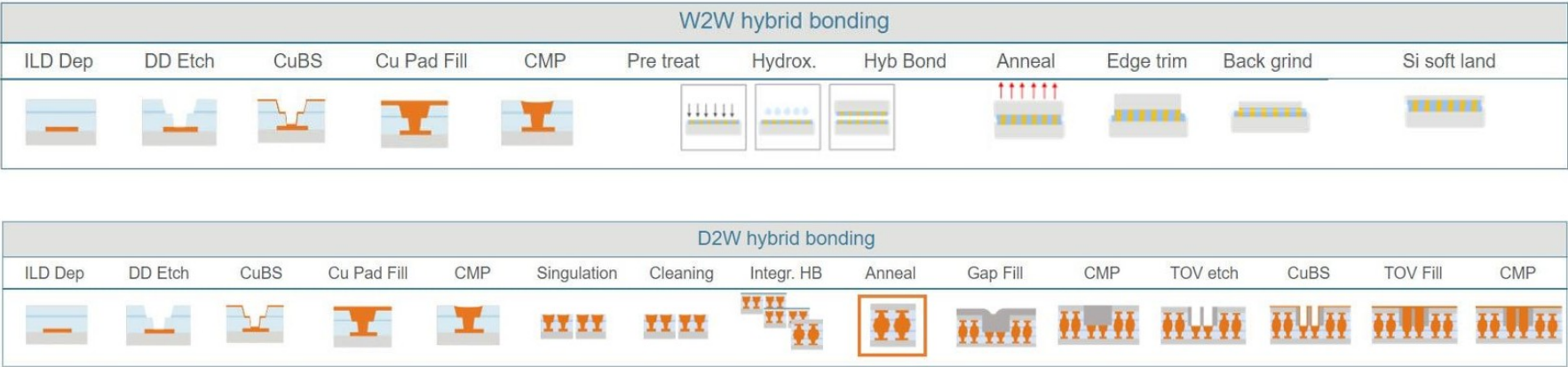
Hybrid Bonding Process

Example applications



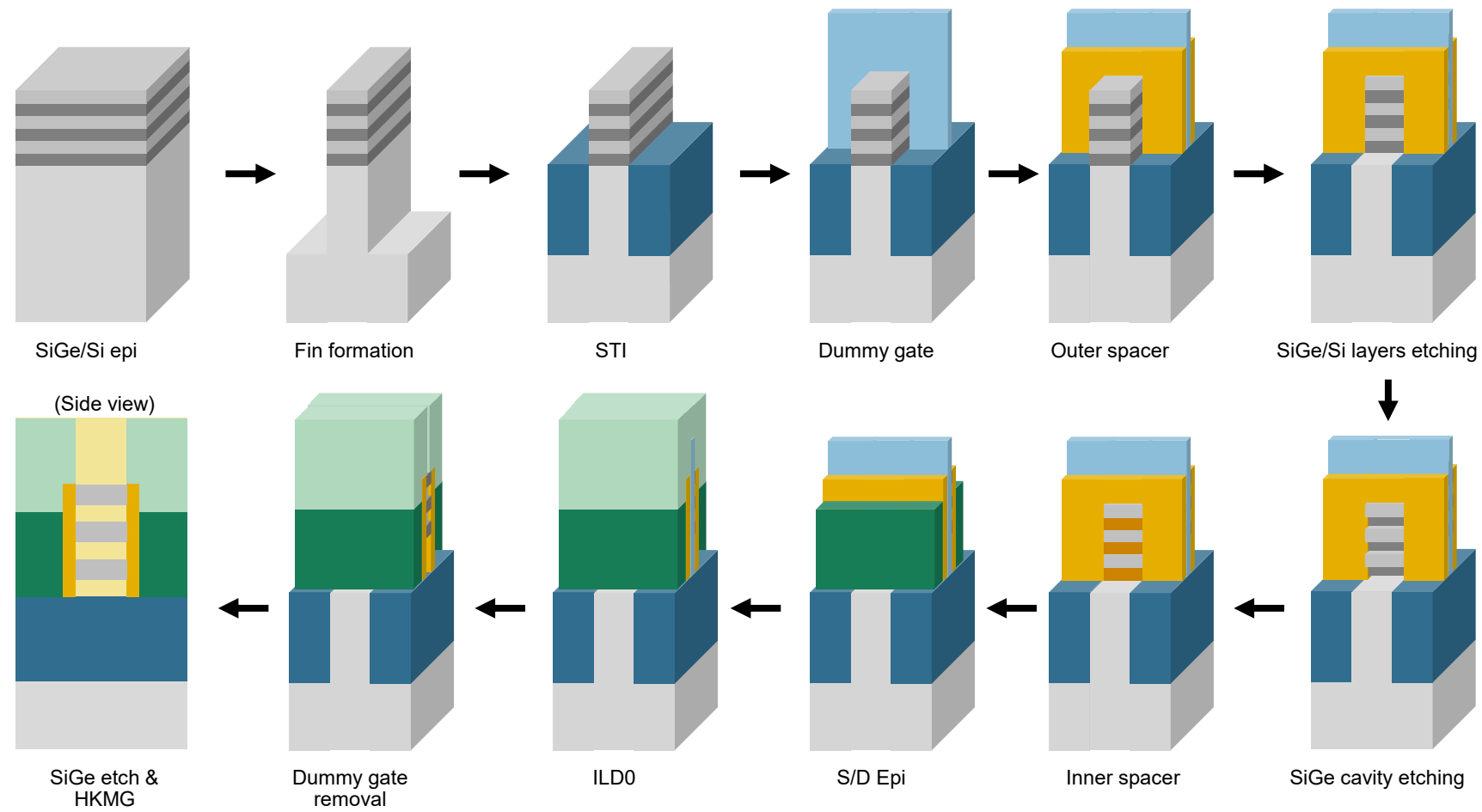
Process

- 1. Surface treatment & cleaning — plasma activation of the wafer surface and cleaning to remove particles
- 2. Alignment — precise positioning so the metals make accurate contact
- 3. Bonding & anneal — press-bonded at room temperature, with heat treatment to increase bond strength



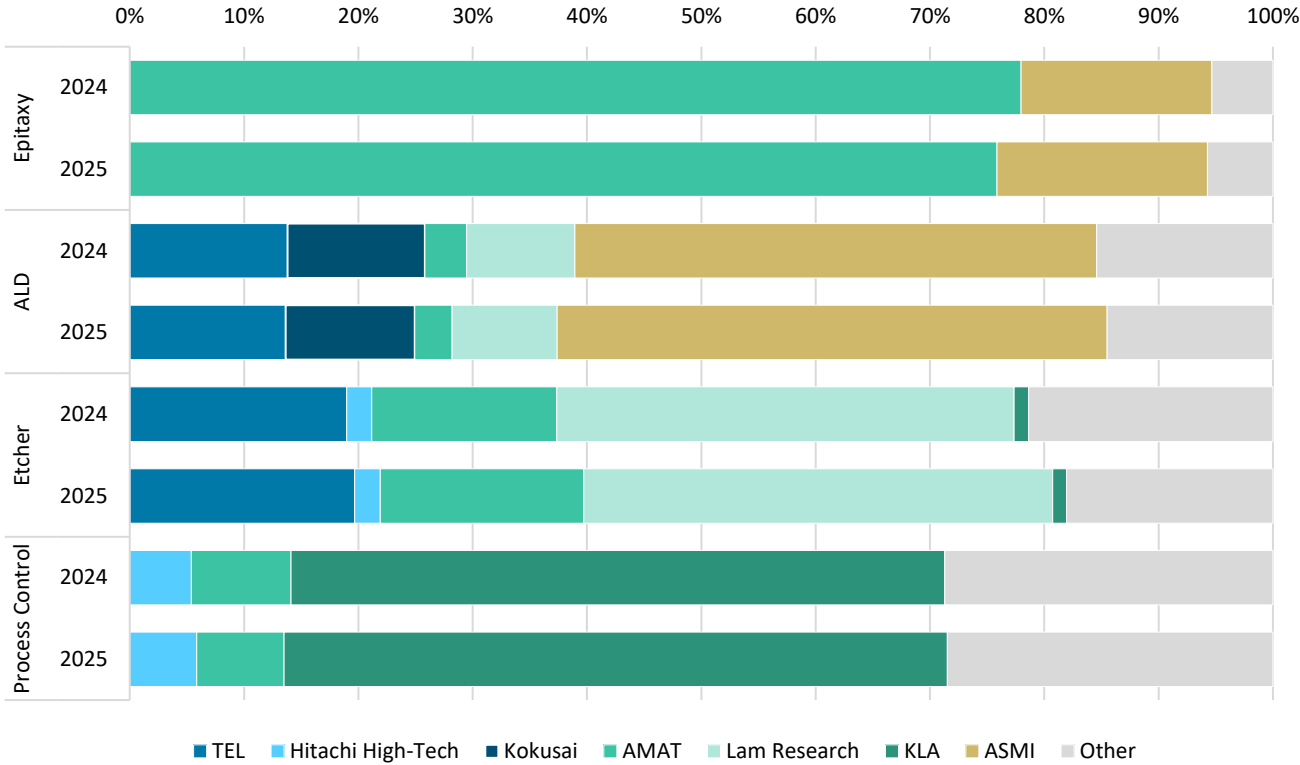
Source: Applied Materials

GAA Process



Source: Morgan Stanley Research

GAA-Related Equipment Share (2025)



Source: Company data, Gartner, Morgan Stanley Research

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(as of June 30, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1544	42%	453	49%	29%	757	44%
Equal-weight/Hold	1577	43%	390	42%	25%	769	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	544	15%	89	10%	16%	204	12%
Total	3,668		933			1731	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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COMPANY (TICKER)	RATING (AS OF)	PRICE* (07/16/2026)
Suzune Tamura, CFA		
Advantest (6857.T)	O (01/17/2024)	¥29,640
DISCO (6146.T)	O (07/07/2023)	¥70,590
Ebara (6361.T)	O (12/04/2024)	¥5,830
JEOL (6951.T)	E (07/08/2025)	¥8,660
KOKUSAI ELECTRIC (6525.T)	O (12/19/2025)	¥9,890
Lasertec (6920.T)	U (07/28/2025)	¥46,370
Nikon (7731.T)	U (08/08/2023)	¥2,298
SCREEN Holdings (7735.T)	O (11/13/2025)	¥18,770
Tekscend Photomask (429A.T)	E (11/18/2025)	¥3,900

Tokyo Electron (8035.T)	O (12/19/2025)	¥70,890
Ulvac (6728.T)	E (01/17/2024)	¥10,305
Ushio (6925.T)	O (01/05/2026)	¥4,076

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